

TANAY SINGH CHAUHAN

EDUCATION

Indian Institute of Technology (BHU), Varanasi	2024 - 2028
Bachelor of Technology in Ceramic Engineering	7.94
Sanskar Public School	2024
CBSE	90.80%
St Pauls School	2022
CBSE	87.20%

SKILLS

- **Programming Languages:** C++, Python, SQL, TypeScript, JavaScript, HTML, CSS
- **Frameworks & Libraries:** React.js, Next.js, NestJS, FastAPI, Tailwind CSS, Prisma ORM, SQLAlchemy, Pydantic, Scikit-learn, Pandas, NumPy, TanStack Query, Zustand, Plotly
- **Core Computer Science:** Data Structures & Algorithms, Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems, Computer Networks, REST APIs
- **Machine Learning:** Data Preprocessing, Exploratory Data Analysis (EDA), Feature Engineering, Model Evaluation, Hyperparameter Tuning, Scikit-learn Pipelines
- **Developer Tools:** Git, GitHub, Docker, Turborepo, pnpm Workspaces, Postman, Swagger/OpenAPI, Conda, Uvicorn, Vercel, Railway
- **Databases:** PostgreSQL, SQLite

PROJECTS

- ShareFlow** *Deployed · Github | May 2026*
- Built a secure **file-sharing platform** using NestJS, React, PostgreSQL, and Docker that supports resumable 2GB uploads, password-protected links, and one-time downloads for reliable **large-file sharing**
 - Implemented chunked streaming uploads, SHA-256 content deduplication, and HMAC-signed download URLs to **improve storage efficiency, security, and transfer reliability**.
 - Architected a production-grade backend with background cleanup workers, pluggable cloud storage support, and streaming I/O for scalable, memory-efficient file transfers.
- AlgoVerse** *Deployed · Github | March 2026*
- Built a full-stack **DSA learning platform** using Next.js, NestJS, PostgreSQL, and Prisma, offering structured algorithm explanations, multilingual implementations, and curated practice problems.
 - Developed scalable REST APIs, advanced search, category-based navigation, and persistent progress tracking using TanStack Query and Zustand for a responsive user experience.
 - Engineered a modular **Turborepo monorepo** with shared TypeScript contracts, server-side rendering, and reusable backend architecture for maintainable and scalable development.
- AutoML Studio** *Deployed · Github | December 2025*
- Built a no-code AutoML platform using FastAPI, React, and Scikit-learn that **automates data preprocessing, EDA**, model training, hyperparameter tuning, and predictions from CSV datasets.
 - Developed reusable **machine learning pipelines** with automated preprocessing, GridSearchCV optimization, interactive Plotly dashboards, and downloadable production-ready models.
 - Designed a **modular backend architecture** with session-based model management, dynamic prediction interfaces, and extensible model registries for scalable ML experimentation.

ACHIEVEMENTS

- Achieved a peak **Leetcode** rating of **1755**.
- Achieved a peak **Codeforces** rating of **1342 (Pupil)**.
- Secured **Rank 917** in Codeforces Round 1107 and **Rank 1247** in LeetCode Weekly Contest 509.
- Solved **900+** **algorithmic and data structure problems** across competitive programming platforms.

CERTIFIED COURSES

- *Dynamic Programming Camp* — **AlgoUniversity**; covered Dynamic Programming, Matrix Exponentiation, and Square Root Decomposition.
- *Advanced Learning Algorithms* — **DeepLearning.AI & Stanford Online** (Coursera)
- *Supervised Machine Learning: Regression and Classification* **DeepLearning.AI & Stanford Online** (Coursera)

POSITIONS OF RESPONSIBILITY

Induction Mentor

Guided and supported incoming freshmen during the institute induction program, helping them transition smoothly into college life.